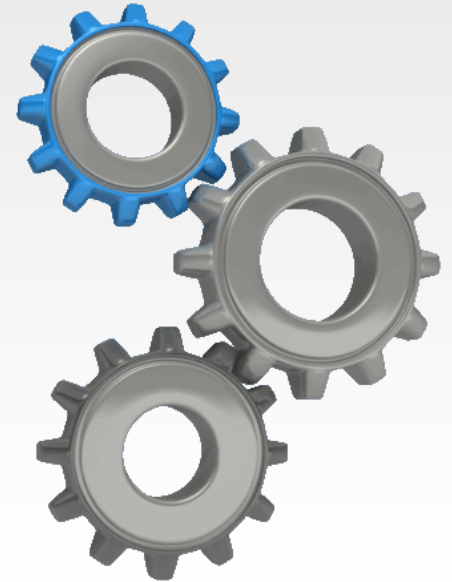


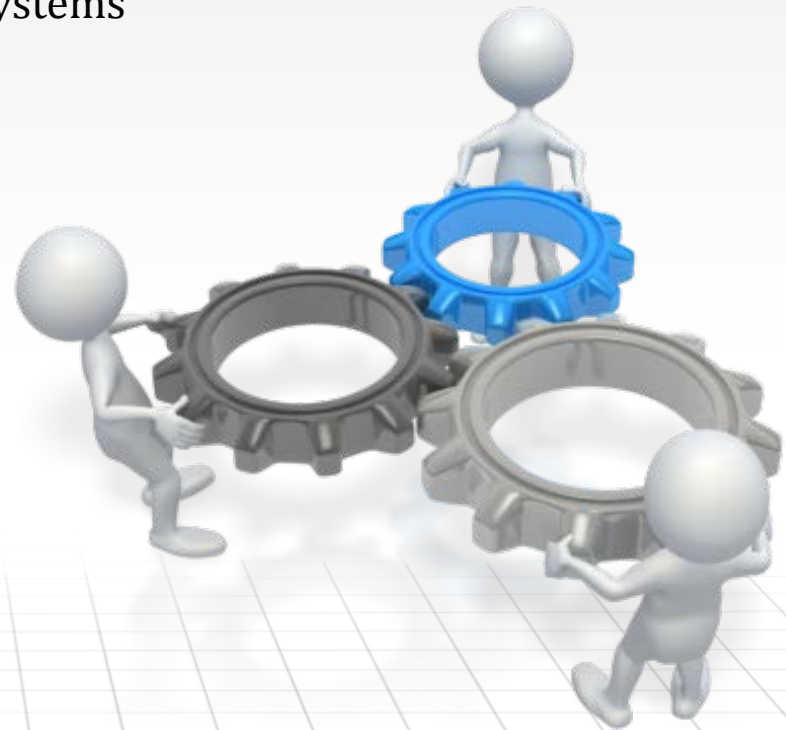


LECTURE 2

MANUFACTURING SYSTEMS

Contents

1. Objectives
2. Review
3. Manufacturing systems
4. Components of a manufacturing system
5. Types of manufacturing systems
6. Classification of manufacturing systems
7. Manufacturing progress function
8. Conclusion



Objectives

1. Define the types of manufacturing systems
2. Outline the classification criteria for manufacturing systems
3. Understand the learning curve



Review

1. The role of manufacturing
2. Integrated manufacturing systems
3. The factors that characterize a manufacturing system
4. The components of a CIM system



Manufacturing system (MS)

A manufacturing system is a collection of integrated equipment and human resources, whose function is to perform one or more processing and/or assembly operations on a starting raw material, part, or set of parts.

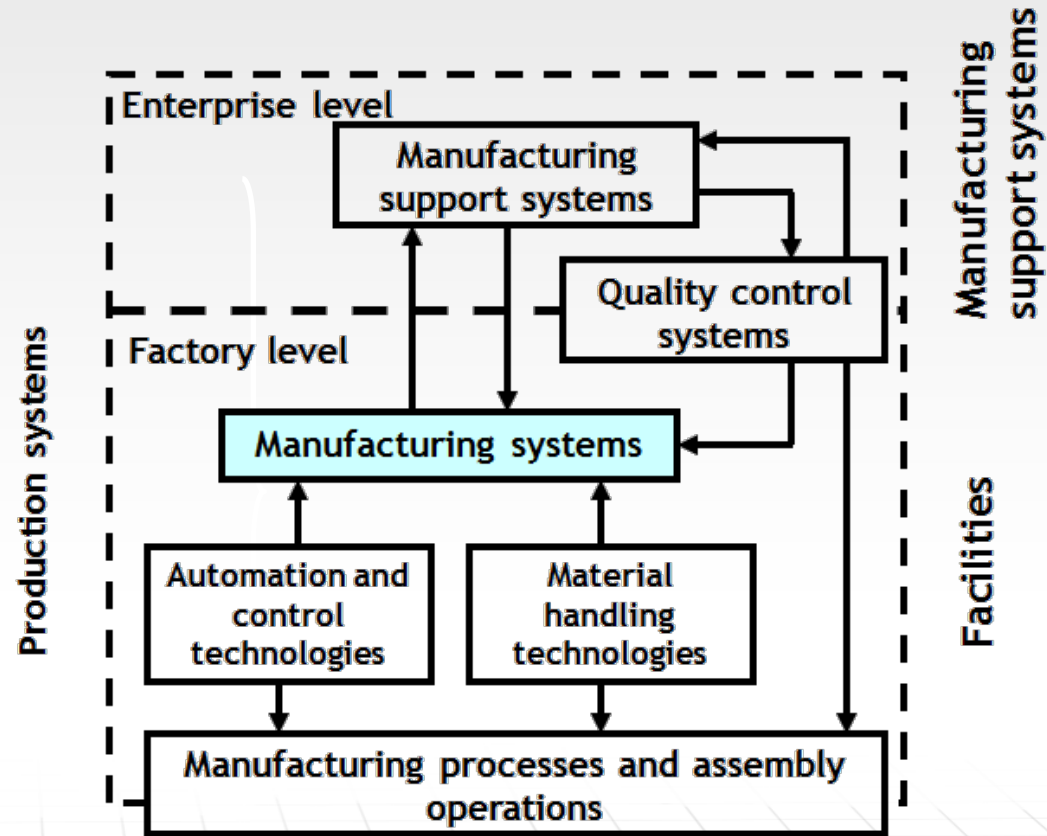


Figure 2.1 The position of the manufacturing system in the larger production system

Components of a MS

- Production machines plus tools, and other related hardware
- Material handling system
- Computer system used for control
- Human workers





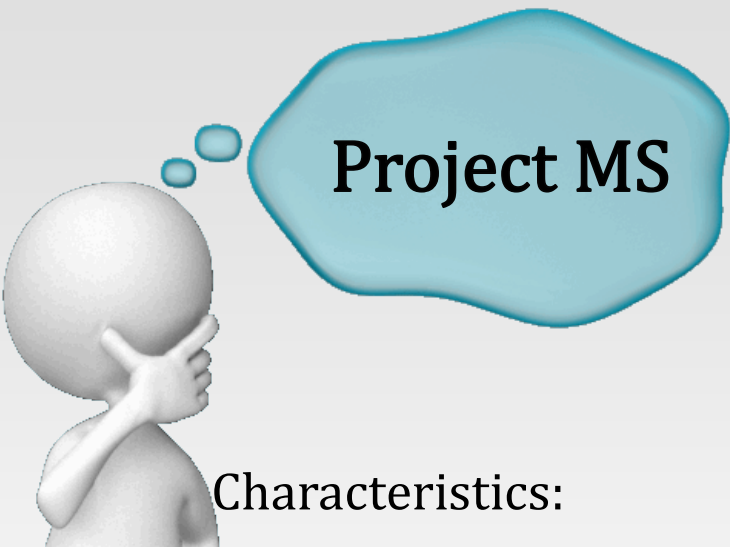
Types of MS

Types of manufacturing systems by production operations:

PROJECT
JOB SHOP
REPETITIVE
LINE
CONTINUOUS



Classification of companies into these groups requires a detailed analysis and evaluation of the production operations.



Project MS

Characteristics:

- **Complex** products.
- The plant layout is called **fixed position**.
- The **design** drawings are **complex**.
- The **lead time is long**.
- The **customer is identified before** production starts.
- Production scheduling usually uses project management techniques.



Project-type companies: build oil refineries, large office buildings, cruise ships, large aircraft etc.

Job shop MS

Characteristics:

- Less than **20%** repeat production on the **same part**.
- **Noncomplex** products.
- **Intensive scheduling and routing** on the shop floor.
- The **raw material purchased as needed** for each project.
- **Low volume and productions quantities**, called lot sizes, that are small.

Office	Milling	Lathe	Grinding	Heat treating
	Assembly	Finishing	Drilling	Shopping and receiving

Figure 2.2 Process layout



The classic machine shop with lathes, mills, grinders, and drill press is included in this category.



Repetitive MS

Characteristics:

- Orders for **repeat business** approach 100 percent,
- Orders have a **moderately high volume**, with **lot sizes varying over a wide range**.
- Blanket **contracts** with customers **for multiple years** occur frequently.
- Orders have **fixed routings** for the production machines.
- The **plant layout** could be either *the process layout* (Figure 2.2) or more like *the product-flow layout* (Figure 2.3).

The production machines are frequently special purpose machines, called *transfer machines* build to product a specific product or a family of products.

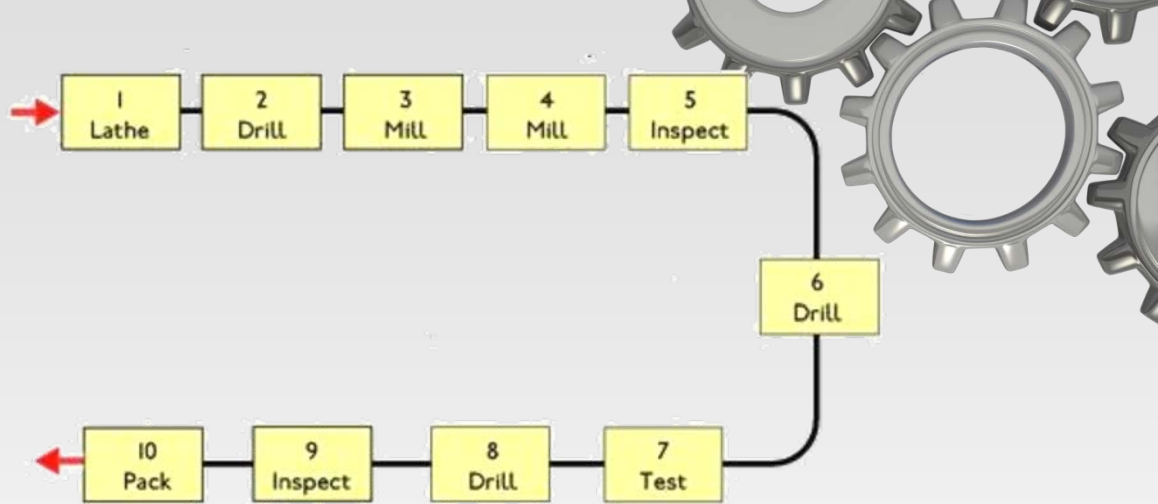
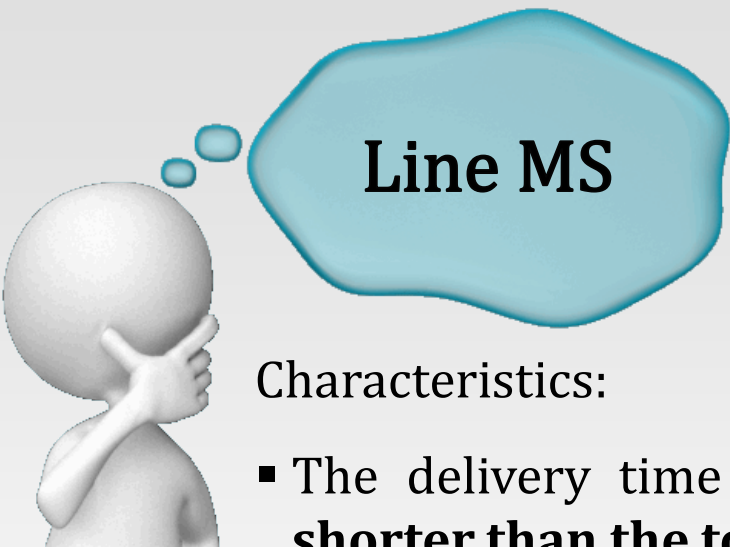


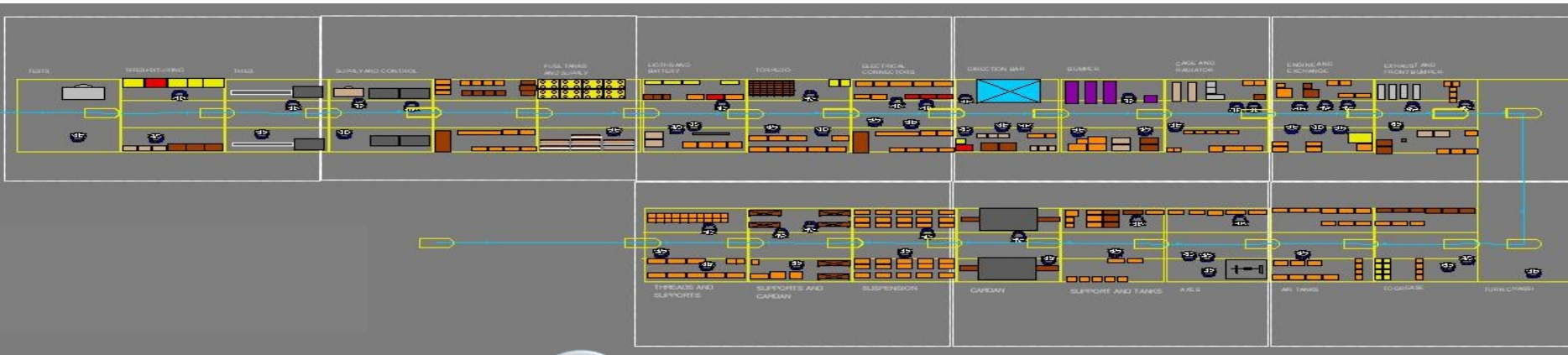
Figure 2.3 Product flow layout

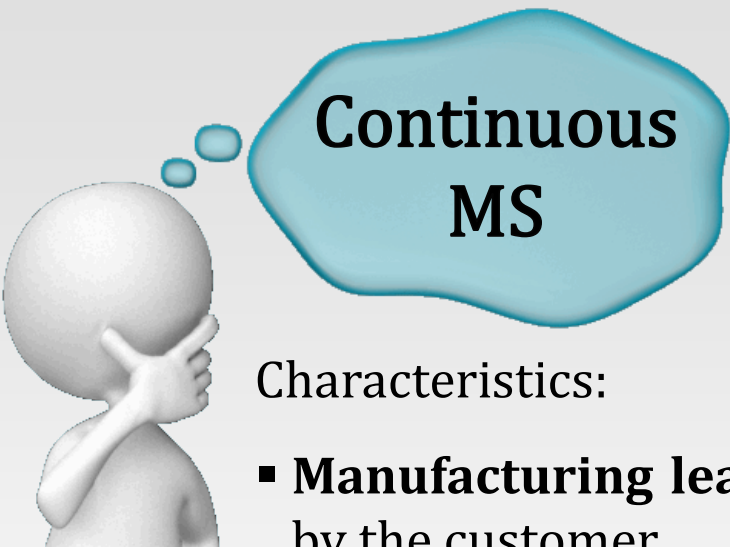


Line MS

Characteristics:

- The delivery time (**lead time**) required by the customer is often **shorter than the total time** it takes to build the product.
- The **product has many different options or models**.
- **An inventory of subassemblies is normally present.**
- The **plant layout** usually used is the **product-flow layout** (Figure 2.3).





Continuous MS

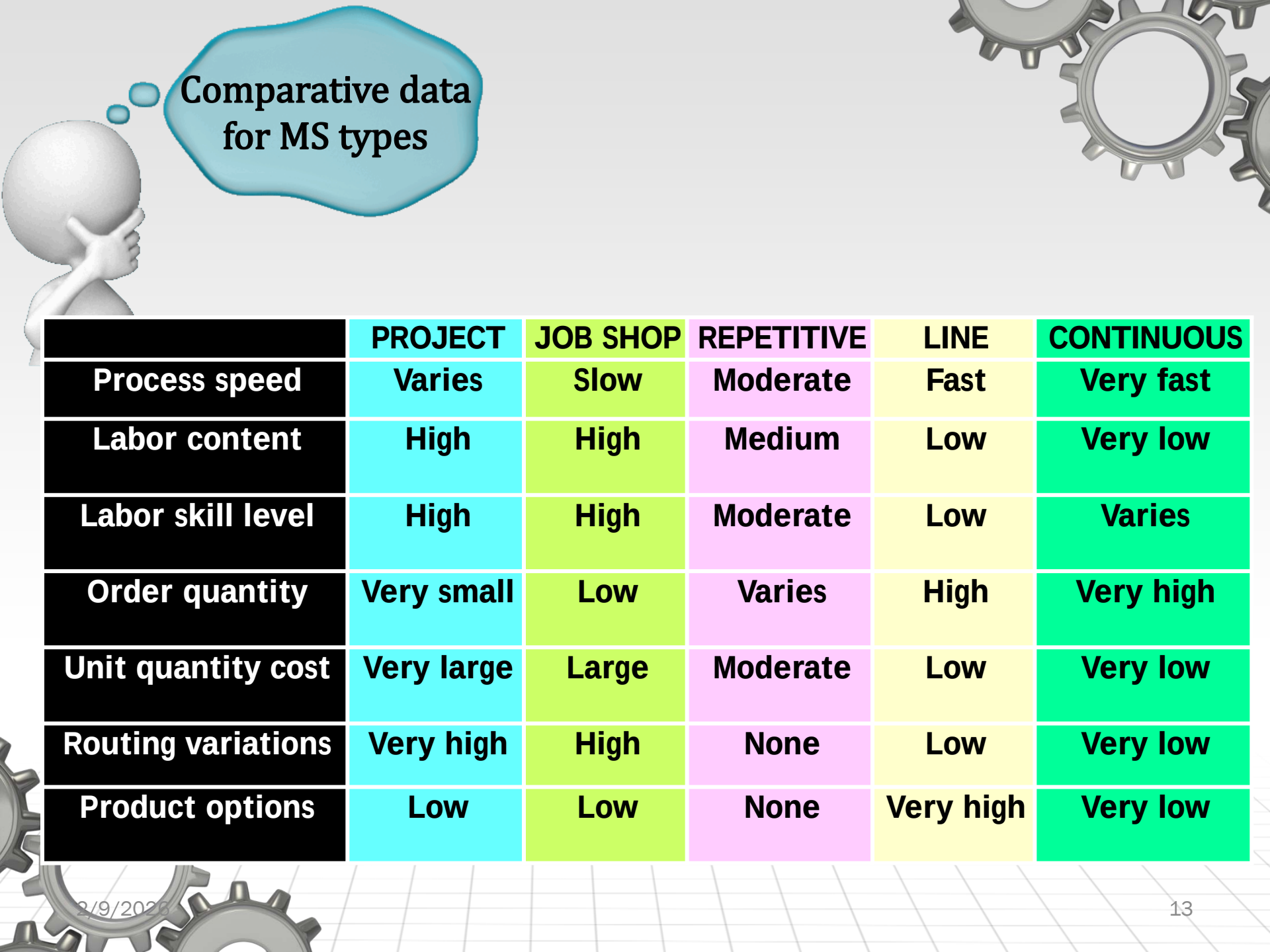
Characteristics:

- **Manufacturing lead time is greater than the lead time expected** by the customer.
- **Product demand is predictable.**
- **An inventory** of finished products is maintained.
- **Production volume is high.**
- **Products have few options.**
- The **plant layout** always used is the ***product-flow layout*** (Figure 2.3).



Examples of this industry type include the production of breakfast cereal, the production of petroleum and chemical products etc.

**Comparative data
for MS types**



	PROJECT	JOB SHOP	REPETITIVE	LINE	CONTINUOUS
Process speed	Varies	Slow	Moderate	Fast	Very fast
Labor content	High	High	Medium	Low	Very low
Labor skill level	High	High	Moderate	Low	Varies
Order quantity	Very small	Low	Varies	High	Very high
Unit quantity cost	Very large	Large	Moderate	Low	Very low
Routing variations	Very high	High	None	Low	Very low
Product options	Low	Low	None	Very high	Very low



Self-assessment test

1. In any production system the human resource is present.

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2. A rocket requires a job shop-type manufacturing system.

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3. Bread production requires a job shop-type manufacturing system.

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4. A car requires a line manufacturing system.

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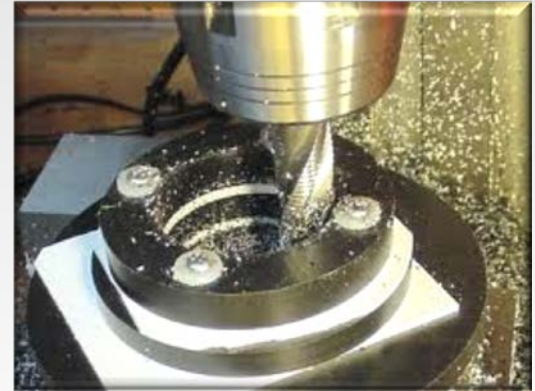
Classification of MS



Factor	Alternatives
(1) Types of operations performed	Processing operations versus assembly operations. Type of processing or assembly operation.
(2) Number of workstations and system layout	One station versus more than one station. For more than one station, variable routing versus fixed routing.
(3) Level of automation	Manual or semi-automated workstations that require full-time operator attention versus fully automated that require only periodic worker attention.
(4) Part or product variety	All work units identical versus variations in work units that require differences in processing.

Types of operations performed

- At the highest level, there are two types of operations:
 - Processing operations
 - Assembly operations
- Additional parameters of the product that play a role in determining the design of the manufacturing system include:
 - type of material processed
 - size and weight of the part or product
 - part geometry



Number of workstations and system layout

- The number of workstations in the manufacturing system is a convenient measure of its size.
- As the number of stations is increased, the amount of work that can be accomplished by the system increases.
- According to number of stations and the layout of the stations there are three levels:
 - Single station,
 - Multiple stations with variable routing,
 - Multiple stations with fixed routing.



Level of automation

- According to this factor, manufacturing systems can be:
 - manually operated,
 - semi-automated,
 - automated.
- The manning level of a workstation, symbolized M , is the proportion of time that a worker is in attendance at the station:

$$M = \frac{w_u + \sum_{i=1}^n w_i}{n} = \frac{w}{n}$$

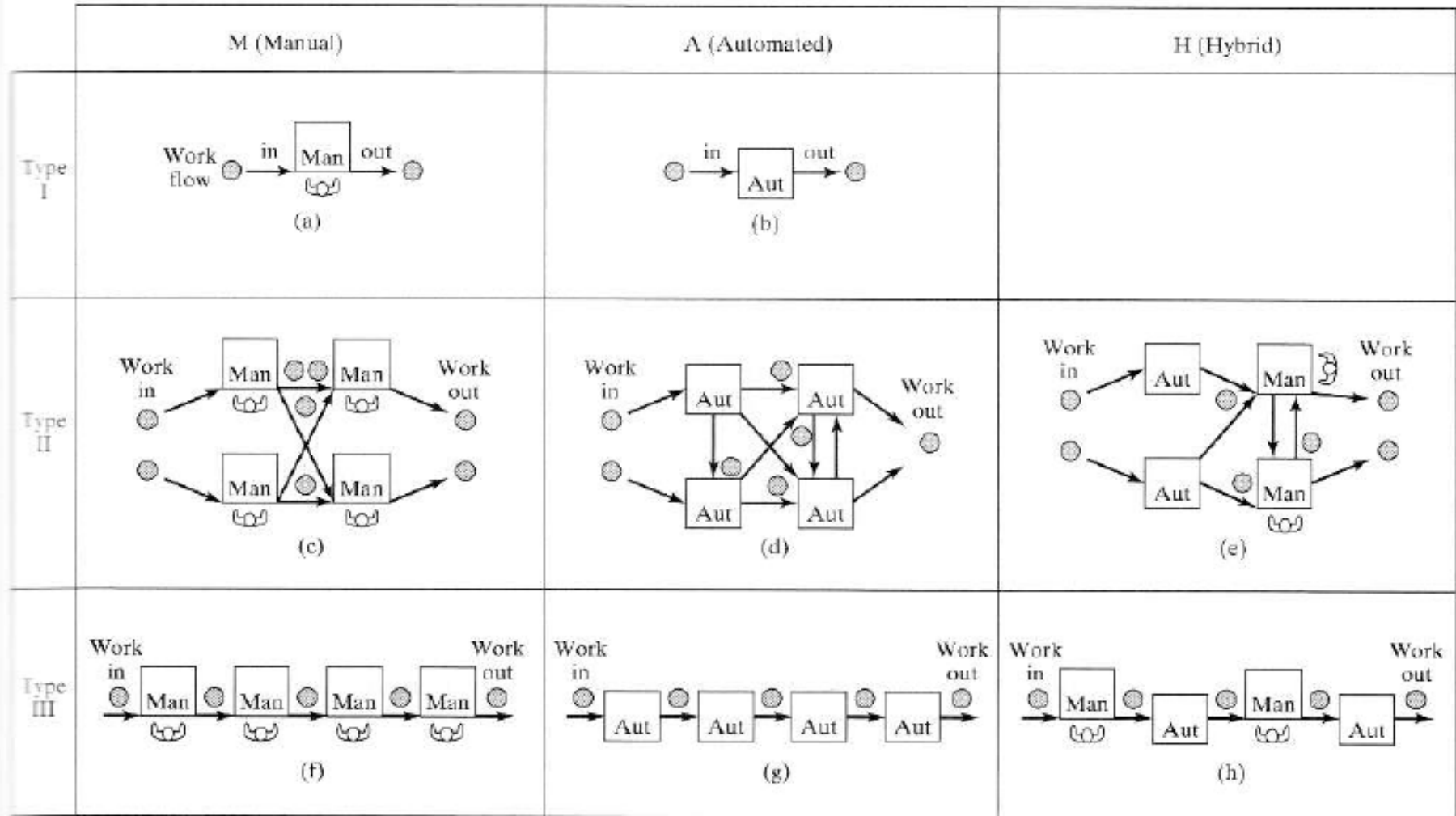
M – average manning level for the system

w_u – number of utility workers assigned to the system

w_i – number of workers assigned specifically to station i for $i=1..n$

Utility workers $\Leftrightarrow$ workers who are not specifically assigned to individual processing or assembly station

Level of automation



Flexibility and productivity

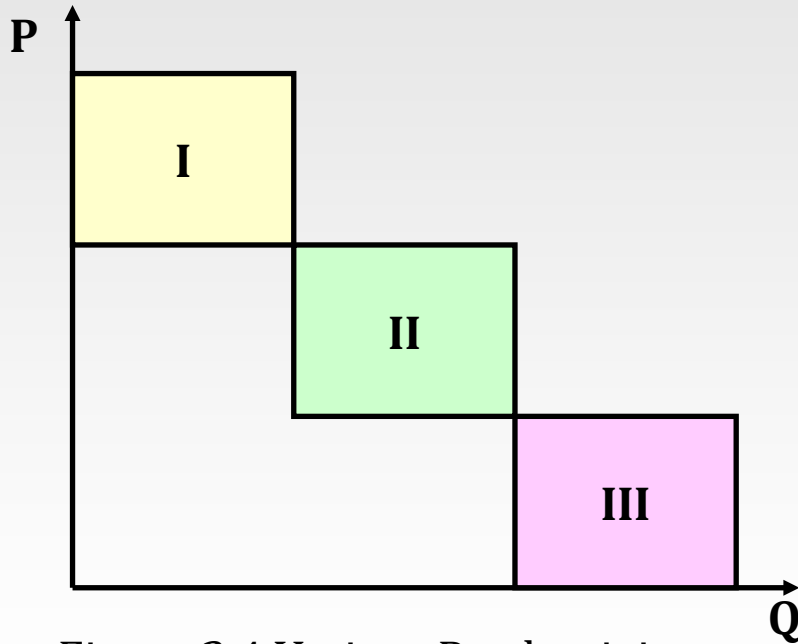


Figure 2.4 Variety-Productivity

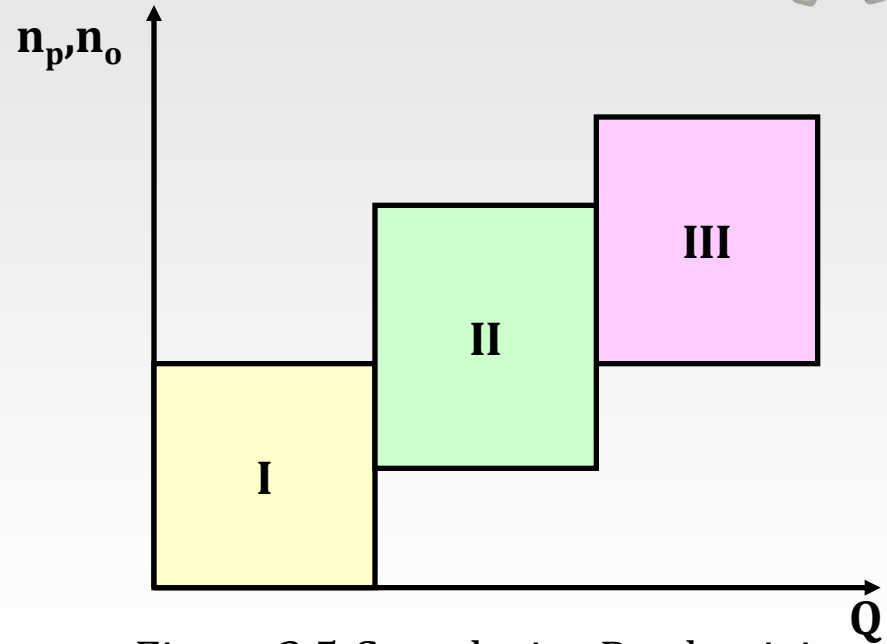
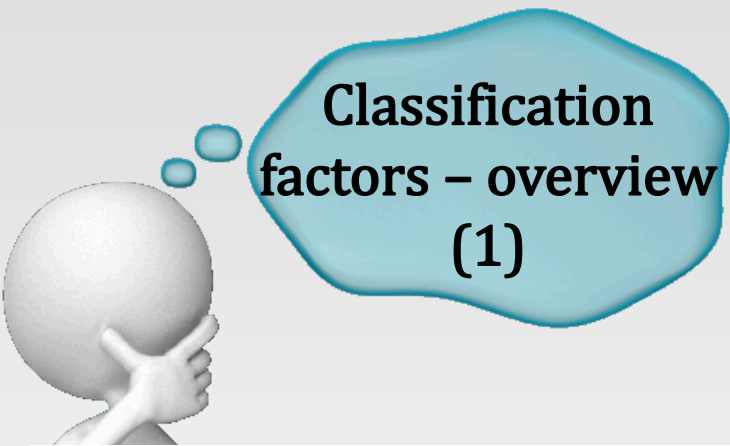


Figure 2.5 Complexity-Productivity

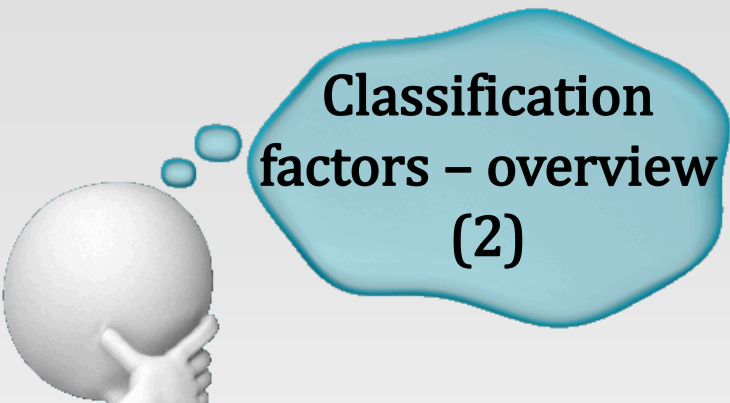
Notations: P – part or product variety; Q – annual production quantity
n_p – number of parts/products; n_o – number of operations
I – Single station; II – Multiple stations with variable routing;
III – Multiple stations with fixed routing



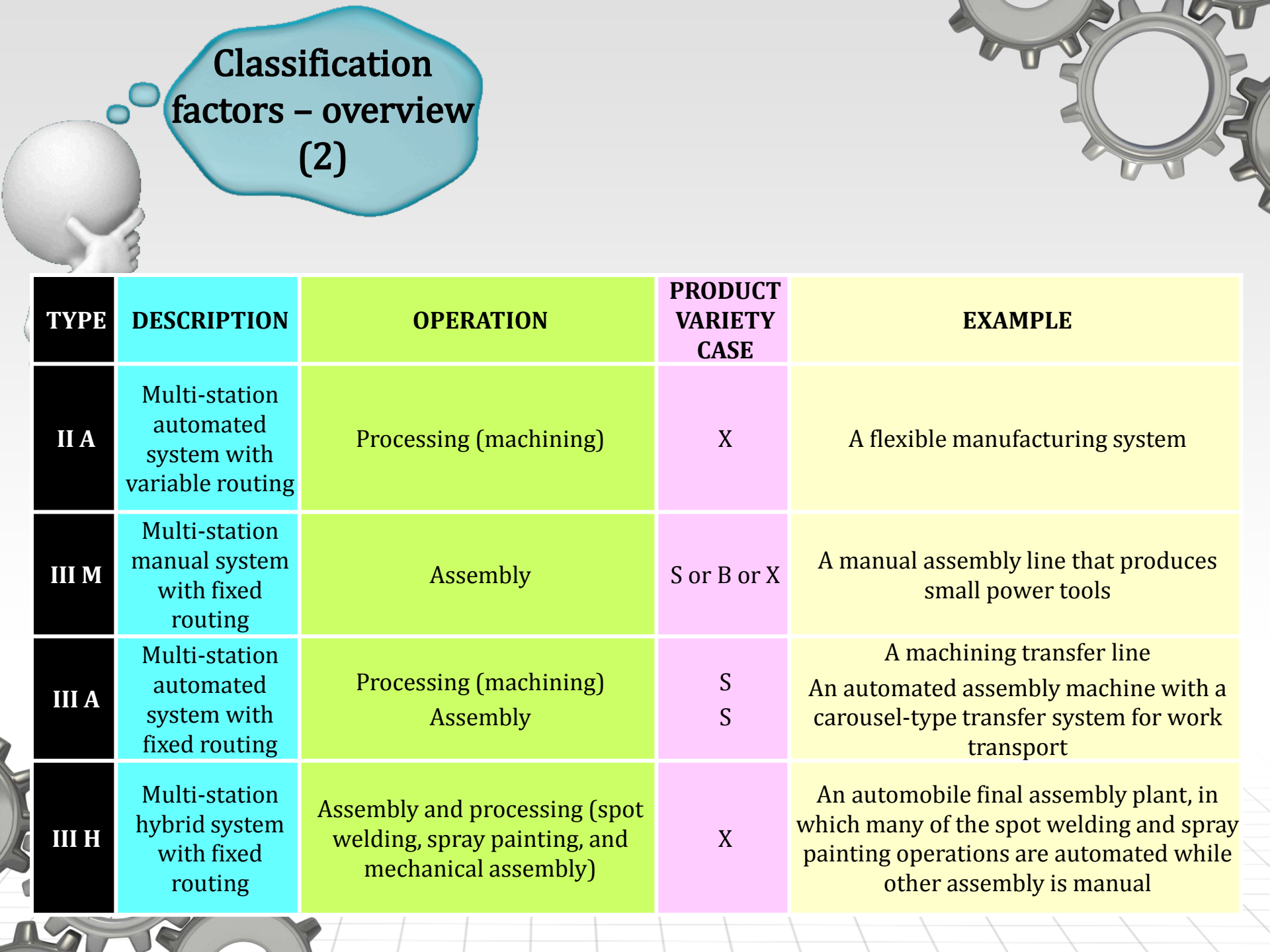
Classification factors – overview (1)



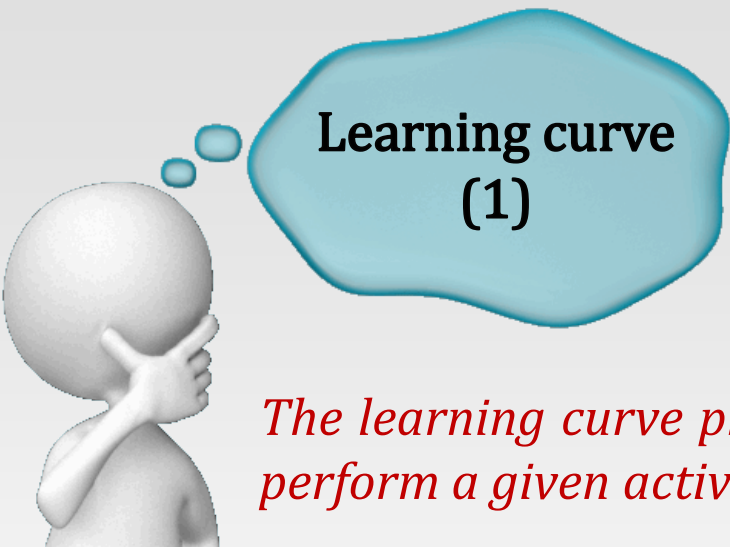
TYPE	DESCRIPTION	OPERATION	PRODUCT VARIETY CASE	EXAMPLE
I M	Single station manned cell	Processing (machining) Processing (stamping) Assembly (welding)	S or B S or B S or B or X	Worker at CNC lathe Worker at stamping press Worker and fitter at arc welding setup
I A	Single station automated cell	Processing (machining) Assembly (mechanical)	B or X S or X	A CNC machining center with parts carousel operating in an unattended mode An assembly system in which one robot performs multiple assembly tasks to complete a product
II M	Multi-station manual system with variable routing	Processing (machining) Processing (machining)	X B	A group technology machine cell that machines a family of metal parts A small job shop with a process layout might be considered II M system. It produces a variety of different part or product styles requiring a variety of process routings



Classification factors – overview (2)



TYPE	DESCRIPTION	OPERATION	PRODUCT VARIETY CASE	EXAMPLE
II A	Multi-station automated system with variable routing	Processing (machining)	X	A flexible manufacturing system
III M	Multi-station manual system with fixed routing	Assembly	S or B or X	A manual assembly line that produces small power tools
III A	Multi-station automated system with fixed routing	Processing (machining) Assembly	S S	A machining transfer line An automated assembly machine with a carousel-type transfer system for work transport
III H	Multi-station hybrid system with fixed routing	Assembly and processing (spot welding, spray painting, and mechanical assembly)	X	An automobile final assembly plant, in which many of the spot welding and spray painting operations are automated while other assembly is manual



Learning curve (1)

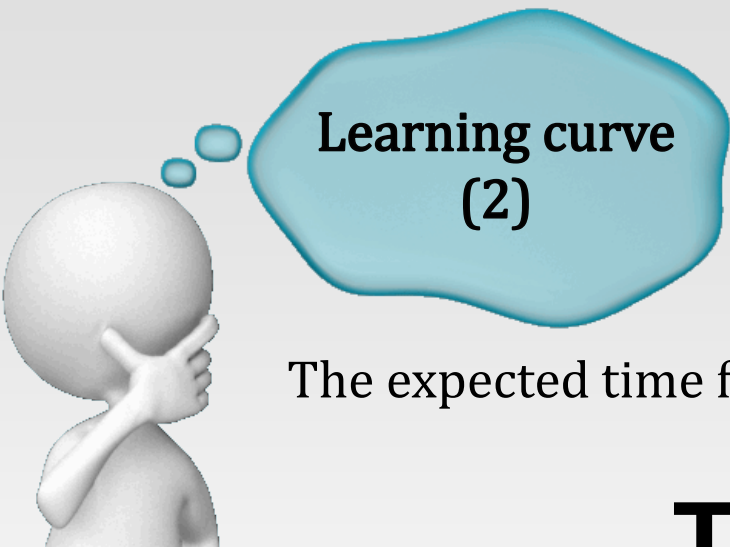
The learning curve phenomenon occurs when the cycle time required to perform a given activity decreases as the number of cycles increases.



In the case of work teams, large organizations, and manufacturing systems the phenomenon is called *manufacturing progress function*.

According to learning curve theory, there is a *constant learning rate* that applies to a given task.

TYPE OF WORK	TYPICAL LEARNING RATE (%)
Assembly, electrical harness	85
Assembly, electronic	85
Assembly, mechanical	84
Assembly of prototypes	65
Inspection	86
Machining	90-95
Sheet metal working	90
Welding	85



Learning curve (2)

The expected time for the N^{th} work unit is:

$$T_N = T_1 \times (N)^m$$

where:

T_N - task time for the N^{th} unit of work,

T_1 - task time for the first work unit;

N - the number of the unit produced in the series

m - an exponent that depends on the learning rate

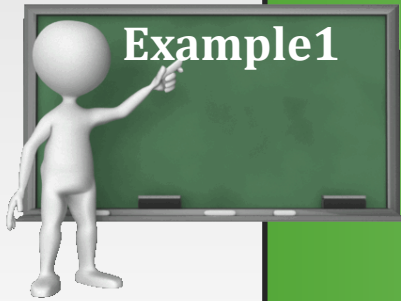
$$m = \frac{\ln(LR)}{\ln(2)}$$

where:

LR - learning rate expressed as a decimal fraction



Learning curve (manufacturing progress functions)



A certain mechanical assembly task require 3.75 min to complete when a skilled worker did it for the first time. The task will be performed on an assembly line used to produce 1000 units of a particular product. The line is currently operating on a pilot basis, while workers are learning their respective tasks. The line will run on this basis for 50 units, then it will go into regular production.

- a). If the learning rate for tasks of this type is 84% what will the task time be for the 50th unit?
- b). But for the 1000th unit?

Solution:

$$m = \frac{\ln(0.84)}{\ln(2)} = -0.2515$$

$$T_{50} = 3.75 \times (50)^{-0.2515} = 3.75 \times (0.3738) = 1.402 \text{ min}$$

$$T_{1000} = 3.75 \times (1000)^{-0.2515} = 0.660 \text{ min}$$



Self-assessment test



1. In a two workstation manufacturing system, the routings may be fixed or variable.

T

F

2. In a production system with more than two workstations, the routings can be fixed or variable.

T

F

3. A flexible manufacturing system is a system with one automated workstation with variable routings.

T

F

4. A flexible manufacturing system is a system with multiple automated workstation with fixed routings.

T

F

5. The learning curve occurs when the time to perform a given operation decreases with the number of repetitions of the same operation.

T

F



Conclusion



- ☐ The complexity of a product and the required quantity determine the choice of the appropriate manufacturing system.
- ☐ The level of automation of a manufacturing system determines the importance of human resources.
- ☐ The performance of a manufacturing system can be analyzed also in terms of progress function (learning curve).





- ☐ Manufacturing operations
- ☐ Product – production relationship
- ☐ Production concepts used in manufacturing
- ☐ Manufacturing cost and selling price
- ☐ Production strategies